Measuring Generalization Surprise in AI

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Abstract

To establish reliable expectations and minimize surprise for users and policymakers, an AI system's reported evaluation suite should accurately characterize system behavior and capabilities. This remains challenging because AI systems generalize in unpredictable ways and popular evaluation suites overlook human surprise. To address these challenges, we introduce *generalization surprise*, a novel metric measuring how much people would be surprised by an AI system's performance on a new problem, given its previous evaluation results. We operationalize this notion of surprise using a generalization model of human expectations and evaluate it over five families of AI systems on a newly introduced suite of 85 diverse benchmarks. We discover that, despite scoring comparably on standard benchmarks, the latest AI systems can differ markedly in generalization surprise. Even closed frontier systems exhibit myriad generalization failures, especially in tasks outside formal sciences. Our generalization surprise metric can help address this by scalably identifying surprising failures and successes, valuable for smoothing out generalization irregularities and enhancing evaluation construction. Further, unlike conventional leaderboards, we can combine both system error and surprise into a *generalization loss* metric that resists score-hacking (e.g., training on test data) that inflate performance without improving generalization. Additionally, our framework reveals systematic gaps in existing model cards of frontier systems that

limit people's ability to form reliable expectations about AI's generalization, for which we propose improvements. Our conceptual, methodological, and empirical contributions represent a major advance in AI evaluation and data-centric training toward more generalizable and safe general intelligence.

1 Introduction

Proper evaluations have the potential to empower users, policymakers, and researchers to better understand the latest frontier systems, influence AI's trajectory, and help shape society for the better [1–4]. Due to the high stakes associated with frontier AI systems, it has become an industry standard to accompany their releases with associated evaluation suites such as model cards—each typically consisting of a set of benchmarks labeled with aggregate scores such as accuracy and Elo scores [5–8]. Paradoxically, despite their intended role in informing users and policymakers, current evaluations fail to shield humans from myriad surprises or unexpected behavior [9–17], forcing many to rely on ad-hoc out-of-distribution "vibe checks" to superficially intuit the capabilities of AI systems [18].

We attribute this disconnect to two unsolved, intertwined challenges. The first challenge is the *limited generalizability* of AI systems in failing to robustly transfer their performance from benchmark suites to similar, untested tasks, where people naturally expect transfers from a general intelligence. Achieving this human-centric notion of generalization is vital to reduce undesirable surprises since AI usage during deployments is highly dynamic and diverse, unlike existing evaluations that tend to prioritize subtasks that are most convenient to evaluate and hill-climb within given domains [19–21]. The second challenge is that existing *evaluation suites overlook the importance of reducing surprises* when releasing new frontier AI systems. For example, when an evaluation is biased toward certain domains, they provide limited information about how the system will behave in untested use cases encountered during deployment, specially for those use cases that humans perceive as unrelated to the evaluated domains.

Addressing these two challenges is critical for *patching surprising generalization failures* and ensuring that humans can *reliably form expectations about AI behavior*, fostering a broader positive real-world impact of AI. Yet doing so has been difficult. The field still lacks theories and scalable measurement frameworks for evaluating AI's generalization and using such measurements to systematically guide AI training. The current practice measuring generalization is to rely on a set of privately held-out "internal benchmarks," whose test data are explicitly excluded from the training corpus [22–24]. However, as frontier sytems are trained on increasingly diverse training corpora and environments, it has become much harder to define the boundaries between in-distribution and out-of-distribution since training data may contain highly similar or semantically equivalent instances of held-out evaluation tasks [25]; this limits scalability of these internal evaluations. Moreover, to the best of our knowledge, no established measurement tool exists for characterizing how an evaluation suite shapes human expectations about AI behavior nor the extent to which these expectations are broken or fulfilled in novel situations.

To address these issues, we introduce a formal framework for measuring *generalization surprise*, building on a *generalization model* using psychometric features that encodes human expectations of an AI's performance, given knowledge of that system's performance on a previous evaluation suite. This enables us to define (i) metrics for characterizing generalization of AI systems at scale; (ii) a fully-automated methodology for discovering surprising generalization failures in AI; and (iii) a diagnostic tool for assessing the ability of common evaluation suites to properly establish expectations for humans about AI's generalization. We analyze five families of frontier LLMs (GPT, Gemini, Qwen, Muse, Gemma) over a new evaluation suite that we introduce, termed 'SurprisalBench,' covering 538 tasks from 85 benchmarks spanning a wide range of domains.

Our main findings are fourfold. First, we find that recent AI systems, though scoring similarly on standard benchmarks, can diverge notably in generalization surprise. Even closed frontier systems (e.g., GPT-6-Astra) show substantial generalization surprise, especially for tasks outside the domain of quantitative reasoning, such as metacognitive assessment and written expression. Second, unlike standard measures that are commonly used on performance leaderboards, if we combine our notion of surprise with system error into a *generalization loss* metric, we can characterize an AI system's generalization while being resistant to myopic adversarial hill-climbing (e.g., training on test data) that inflate performance on individual benchmarks without improving generalization. Third, the trained generalization models can automatically surface many high-value and low-value items from existing benchmarks where system subjects surprisingly fail and succeed, respectively. Valuing benchmark items using our notion of surprise allows AI developers to systematically identify and patch generalization failures in frontier AI systems, guide the creation of high-value future evaluations, and debug benchmarks (e.g., prompts missing critical information, erroneous gold labels, and brittle programmatic grading). Fourth, model-card evaluations of the latest frontier systems generally lead to substantially more surprises than our proposed approach when extrapolating model performance to unseen tasks, partly due to systematic coverage gaps. We encourage future AI developers to adopt our proposed metrics and methods for improving the training and evaluation suites of AI systems.

2 Modelling Generalization Surprise

Below we detail how we leverage a psychological latent space to capture the human-perceivable features of task items in a manner that is generalizable across arbitrary domains and items. Trained to approximate human generalization expectations given an AI system's past evaluation results, a generalization model takes each new item's latent vector as input to predict how well that AI system will perform on that item. This enables us to measure generalization at scale.

2.1 Properties of Human-centric Generalization

Ref. 26 proposed the universal law of generalization: a psychological space can be established for any set of stimuli (e.g., colors, shapes, sounds) by determining distances

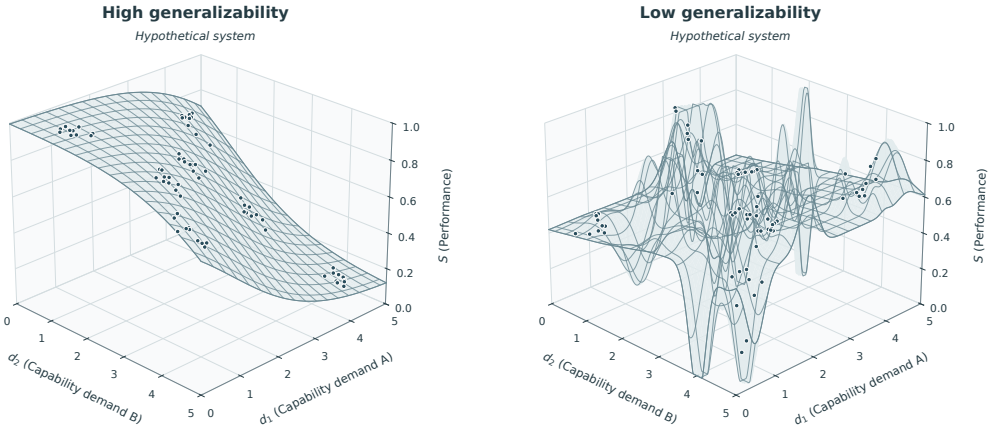


Fig. 1 Intuition of human-centric generalization. We show what high and low human-centric generalizability may look like in a hypothetical 2-D latent space $\mathcal{X} \in [0,5]^2$, where dimensions reflect demand for specific capabilities and dots correspond to task items. (Left) A hypothetical system with high generalizability that exhibits a smooth performance surface. (Right) A system with low generalizability, characterized by a spiky performance surface where negligible shifts in demand result in unpredictable performance swings.

between the stimuli such that the probability that a response learned to any stimulus will generalize to any other is a monotonic function of the distance.

Inspired by Shepard's law, we assume the existence of a latent continuous metric space, $\mathcal{X} \subset \mathbb{R}^K$, with K dimensions that capture *human-perceived generalization patterns*. These K latent dimensions correspond to a set of *general* capabilities (e.g., quantitative reasoning, verbal comprehension, knowledge for formal sciences) required to solve arbitrary cognitive tasks. That is, given a latent space $\mathcal{X}$, any discrete task item i may be characterized by an *item-level demand vector* $\mathbf{x}_i \in \mathcal{X}$.

To define notions of human-centric generalizability within the latent space, we lean on two branches of literature related to Shepard's law of generalization. One branch comes from the machine learning community's notions of robustness [27–30] while another branch draws from item response theory studied in psychometrics [31, 32]. Unifying these two literatures under Shepard's law gives us two properties of human-centric generalization: (1) if two items are close in some latent space $\mathcal{X}$, an AI system's performance on those items should be similar; and (2) if an item is easier than another item, a subject should score higher on the easier item. For intuition, Figure 1 visualizes the two generalization properties for an AI system with high human-centric generalization and one without. For a more formal and extensive treatment of human-centric generalization, see **??**.

2.2 Model of Human-centric Generalization

For simplicity, we implement our generalization model as a binary classifier, such that $S(M,i) \in \{0,1\}$ where 1 indicates model success, but our framework generalizes to multi-class scoring functions. To satisfy both properties of human-centric generalization,

the generalization model p_{θ_M} is restricted to a logistic regression classifier with non-positive coefficients. The logistic classifier approximates property (1) while the non-positive coefficients satisfy property (2).

Formally, our generalization model p_{θ_M}, fed with item-level demand vectors $\mathbf{x}$, is as follows:

$$p_{\theta_M}(y = 1 \mid \mathbf{x}) \;=\; \sigma(\theta_0 + \boldsymbol{\theta}^\top \mathbf{x}), \qquad \boldsymbol{\theta} \in \mathbb{R}^K_{\leq 0}. \tag{1}$$

This sigmoid function follows the standard parametric form used in psychometrics for evaluating human and animal intelligence [33–35], and also recently transferred to AI capability evaluation [4, 32, 36].

2.3 Defining Generalization Surprise

Let $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{ref}}$ denote sets of training and held-out reference items, respectively. Below, j and i correspond to item objects (i.e., prompts in natural language) within $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{ref}}$, respectively, while $\mathbf{x}_j$ and $\mathbf{x}_i$ denote the item-level demand vectors of j and i. And, $S(M, i)$ corresponds to the AI system M's performance score on item i. With this, we define the history of item scores for system M be

$$\mathcal{H}_M = \{(\mathbf{x}_j, S(M, j)) \mid j \in \mathcal{D}_{\text{train}}\} \tag{2}$$

We train the generalization model p_{θ_M} to predict how AI system M performs on each item from $\mathcal{D}_{\text{train}}$ by giving the history $\mathcal{H}_M$ as input, resulting in a $p_{\widehat{\theta}_M}$ with fitted parameters. Then, each held-out item i has a *generalization surprise* score $\ell_{M,i}$ that reflects the extent to which M's actual performance deviates from the expectation of $p_{\widehat{\theta}_M}$:[1]

$$\ell_{M,i} = S(M, i) - p_{\widehat{\theta}_M}(y = 1 \mid \mathbf{x}_i), \qquad i \in \mathcal{D}_{\text{ref}} \tag{3}$$

This can be useful to support sample-efficient AI training by allowing us to scalably locate high-value and low-value items, where M surprisingly fails and succeeds, respectively. Items with negative $\ell_{M,i}$ (surprising failures) reveal critical generalization gaps that should be prioritized during training, while items with positive $\ell_{M,i}$ (surprising successes) disclose areas where the AI scores surprising higher than the expectation formed by previous evaluations.

Intuitively, if system M generalized perfectly to match human expectations, ℓ_{M_i} would always be zero, but non-zero otherwise. If we aggregate $\ell_{M,i}$ over all test items $i \in \mathcal{D}_{\text{ref}}$, we get a global measure of generalization surprise of M[2] :

$$\text{SURPRISE}(M; \mathcal{D}_{\text{ref}}) = \frac{1}{|\mathcal{D}_{\text{ref}}|} \sum_{i \in \mathcal{D}_{\text{ref}}} \ell^2_{M,i} \tag{4}$$

Note that SURPRISE is distinct from standard performance scores like error rate because reduced (average) error does not necessarily lead to reduced generalization surprise (see Figure 2).

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[1] This can be thought of as similar to the notion of surprisal by Ref. 37.

[2] We square $\ell_{M,i}$ so that SURPRISE equals the Brier score [38], a proper scoring rule quantifying the mean squared gap between the generalization model's predictions and M's actual performance.

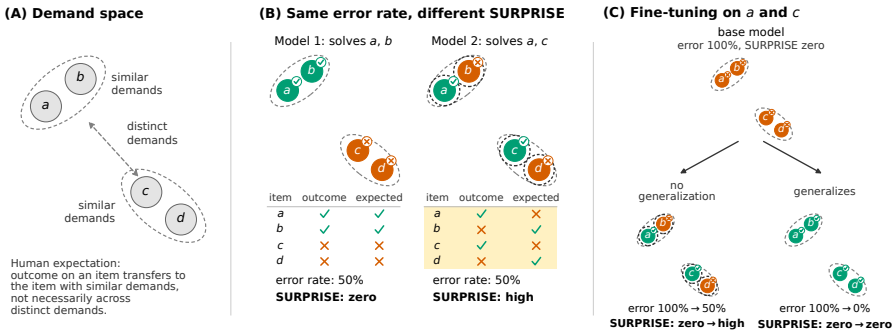


Fig. 2 The difference between generalization surprise and error rate, illustrated in a hypothetical 2D demand space. (A) Four items whose demands form two clusters; a human expects performance to transfer within a cluster ($a \leftrightarrow b$, $c \leftrightarrow d$) but not across clusters. (B) Two AI systems with same error rate but divergent SURPRISE. (C) Fine-tuning a base AI system on a and c reduces the error, but only generalizable fine-tuning keeps SURPRISE at zero.

2.4 Human-centric Generalization Loss

SURPRISE is useful for quantifying overall surprise over $\mathcal{D}_{\text{ref}}$, but it is not equivalent to measuring generalization. For example, when an AI system always fails, its behavior is perfectly predictable and therefore the system has zero SURPRISE (See Figure 2c). To characterize desirable generalization, we instead combine both the AI error and generalization surprise as follows:

$$\text{GENLOSS}(M; \mathcal{D}_{\text{ref}}) = \underbrace{1 - \bar{S}_M}_{\text{error penalty}} + \alpha \underbrace{\left|\bar{S}_M - \bar{P}_M\right|}_{\text{surprise penalty}}, \tag{5}$$

where $\bar{S}_M$ denotes the observed average performance of M over $\mathcal{D}_{\text{ref}}$, and $\bar{P}_M$ denotes its predicted average performance over $\mathcal{D}_{\text{ref}}$. As shown in Figure 3, GENLOSS nullifies gains from above-expected performance (i.e., surprising successes) and penalizes below-expected performance (i.e., generalization failures). Throughout this work, we set $\alpha = 1$. Both penalty terms lie in $[0, 1]$ when $S(\cdot, \cdot)$ is accuracy, so GENLOSS $\in [0, 2]$, with larger values indicating higher error or a larger generalization gap.

GENLOSS represents a *generalizable optimization target*, which can only be minimized if the system improves its performance while transferring such gains in other untrained tasks used to construct the generalization model. To evaluate overall system intelligence, $\mathcal{D}_{\text{ref}}$ should contain diverse cognitive tasks. However, if one wishes to characterize an AI system's generalization for a specific set of items (e.g., in a particular domain like math), they may modify $\mathcal{D}_{\text{ref}}$ to contain a set of items reflective of the set in question that matches their downstream interests.

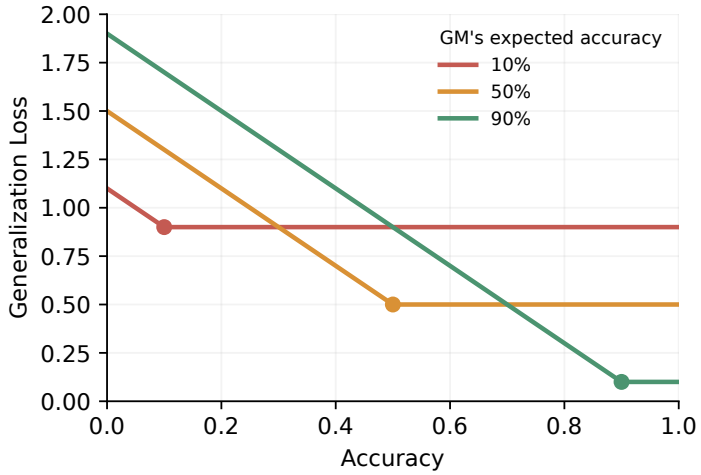


Fig. 3 The connection between generalization loss and accuracy. We plot GENLOSS for three fixed generalization model expectations and show the effects of increasing accuracy under fixed generalization model (GM) expectation. No reward is given for inflated performance and the AI system is penalized when it underperforms the generalization model's expectation.

3 Experimental Design

3.1 Instantiating the Latent Demand Space

Our framework requires instantiating the latent space $\mathcal{X}$. To achieve this, we explore three taxonomies of general cognitive capabilities and skills in the literature: ADeLe v.1.0 [4], O*Net [39] and OECD capability indicators [40]. Appendix **??** elaborates on these three taxonomies and a preliminary large-scale analysis with across 110 latent dimensions. There, ADeLe achieved the best predictive power among the three and thus will be the main taxonomy used in our subsequent analyses. Still, O*Net also performed comparably compared to ADeLe and thus may be considered as an alternative (or complemantary) taxonomy for future work using our framework.

In summary, ADeLe v.1.0 contains 19 dimensions of cognitive capabilities, knowledge and extraneous factors inspired by the literature in human psychology, comparative cognition and artificial intelligence. These include verbal comprehension and expression, logical and quantitative reasoning, mind modeling and social recognition, attention and scan, distinct branches of knowledge, among others. We adopt the subset of 16 dimensions from ADeLe that are supported by literature documenting their presence in humans, while excluding extraneous factors. For simplicity, we refer to this 16-dimensional subset as ADeLe throughout this work. We recommend checking [4] for further details. Each capability dimension is defined by an expert-validated rubric, ranging from 0 to 5—zero represents the absence of capability demand while values 1-5 represent increasing demand levels for that respective capability, with $\mathcal{X} \in [0,5]^{19}$.

3.2 Data Curation and Model Subjects

We collected 85 LLM benchmarks with 538 task categories from major AI conference proceedings, leading scientific journals, and model cards of every studied LLM subject[3]. We exclude agentic tasks that cannot be evaluated on non-agentic LLMs, which leave for future work. We describe the curation details and these benchmarks in Appendix **??**. For improving evaluation efficiency while keeping diversity, we apply medoid-based sampling [42, 43] over item embeddings when curating benchmarks. For each task, we sample up to 50 items by embedding each item with bge-large-en-v1.5 [44], running k-means with $k = \min(50, \text{task size})$, and selecting medoids [42, 43]. This led to a total of 21k items. Collectively, we term this new collection of items "SurprisalBench."

The main pool of subjects comprises 12 LLMs from five families (i.e. GPT, Gemini, Qwen, Gemma, Muse). Six are proprietary systems from OpenAI and Google: GPT-6-Astra, GPT-6.1-Sol, GPT-5.6-Terra, and GPT-6-Luna [6, 45]; and Gemini-4-Argon, Gemini-3.1-Pro and Gemini-3.8-Flash [8, 46, 47]. Other five are open-weight systems from Alibaba, Google, and Meta: Qwen3.6-27B, Qwen3.8-27B, and Qwen3.8-Max [48, 49]; Gemma-4-31B [50]; and Muse-Glimmer-30B [51]. Unless otherwise noticed, our main analyses will focus on these main 12 subjects. Appendix **??** details the inference and grading setups.

3.3 Demand Annotations and Generalization Models

We use GPT-5.6-Terra [45] at low reasoning effort as the demand annotator, using the same prompt and rubrics as Ref. 4 to annotate the demands of items in SurprisalBench. This led to a total of 340K annotations. These annotated demands are fed into the generalization model. Unless otherwise specified, we use leave-one-benchmark-out cross-validation (LOBOCV) across 85 benchmarks for each AI system M. In each fold, we fit a generalization model on items from 84 benchmarks ($\mathcal{D}_{\text{train}}$) and evaluate it on items from the held-out benchmark ($\mathcal{D}_{\text{ref}}$). We then macro-average the resulting held-out results across benchmarks to compute metrics like SURPRISE and GENLOSS. We obtain $R = 1000$ bootstrapping replicates for quantifying epistemic uncertainty. Importantly, through a human study in Appendix **??**, we show that the trained generalization models indeed capture well human-like generalization patterns.

4 Experimental Analyses

4.1 Generalization Surprise

Figure 4 demonstrates that generally the largest close-weight frontier systems not only improve in accuracy but also moderately reduce overall surprise in comparison with earlier counterparts or smaller systems from the same family. In contrast, for less capable AI systems—mostly open-weight systems except for GPT-5.6-Luna—higher accuracy scores do not often lead to lower generalization surprise. We conjecture two complementary explanations: (1) small-sized systems generalize worse and (2) open-weight AI developers are less-resourced and thus more prone to data leakage issues or

[3]All LLM benchmarks of the LLM subjets' model cards are included, with the only exception of CritPt [41] because its private evaluation only returns aggregate metrics and not per-item scores.

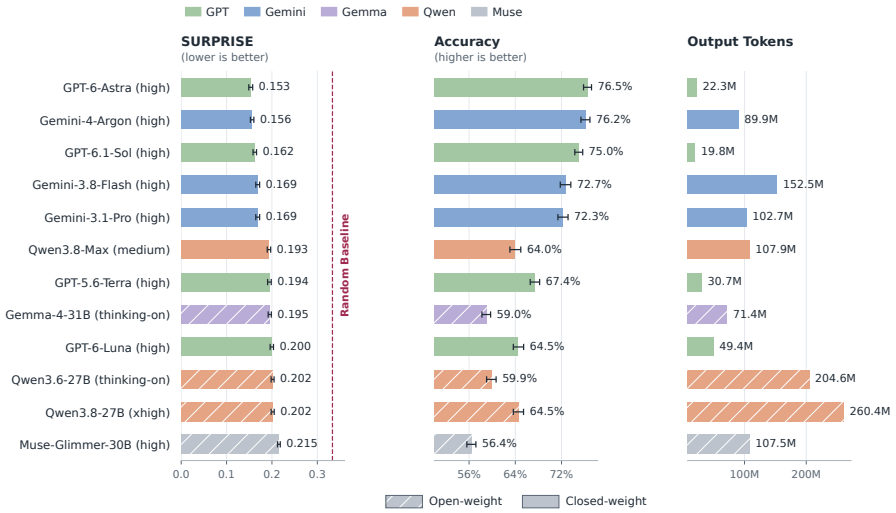


Fig. 4 Generalization surprise and accuracy of studied LLM subjects on our (non-agentic) SurprisalBench. We macro-average SURPRISE under LOBOCV, macro-average accuracy, and total output tokens. Bars are medians and 95% percentile intervals over 1000 bootstrap replicas. Token usage varies across systems, so the rankings should be interpreted with caution.

pressured to train systems on headline benchmarks (i.e., "benchmaxxing") reported by their own model cards [25, 52–54]; Appendix **??** contains additional evidence for these two conjectures.

Despite these improvements, the SURPRISE scores are still large even for the best AI systems relative to a uniform random baseline, indicating substantial room for improvement toward building AI systems that are not only capable but also free from surprising generalization failures.

4.2 Generalization Loss

We characterize model generalization in detail using our generalization loss metric (Methods 2.4) in Figure 5. We observe two key takeaways. First, since GENLOSS is robust against adversarial hill-climbing that inflate performance on individual benchmarks without generalizing to other unseen benchmarks (see Figure 6 and Appendix **??**), the generalization improvements of latest frontier systems seem genuine. Nonetheless, a huge variance between domains still exists. For example, the systems generalize much better in tasks that demand high quantitative reasoning, spatio-physical reasoning, and knowledge in formal sciences (e.g. math, computer science)[4]. In contrast, domains like verbal expression, calibrating knowns and unknowns, and mind modeling manifest much more limited progress, revealing areas that may warrant prioritization in AI training.

These findings can be supported by an further analysis in Appendix **??**, where we show that AI systems tend to reduce their errors without notable reductions in SURPRISE in those (aforementioned) domains with limited progress, unlike other

[4]This seems aligned with the recent focus of frontier labs on math and coding [55–57]

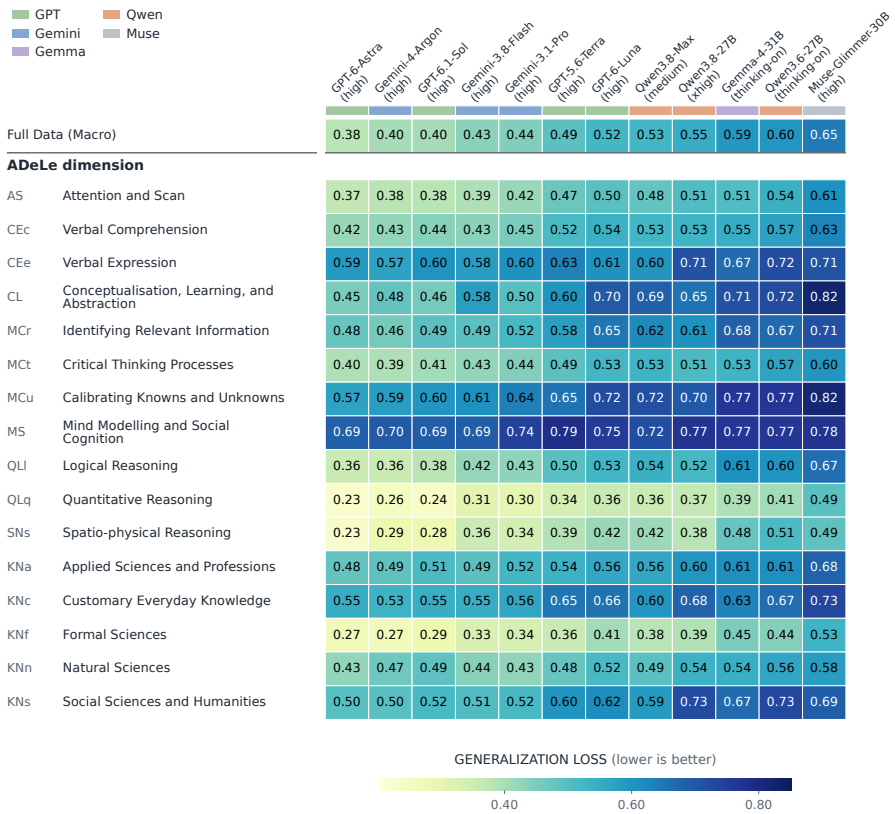


Fig. 5 Generalization loss disaggregated by ADeLe dimensions. The top row is the (total) macro-averaged GENLOSS under LOBOCV. Other domain-specific rows use dominant slicing following [4]: When evaluating a given domain or dimension, keep the items whose demand on that dimension is the highest among other dimensions. The cell numbers are still macro-average across benchmarks, but the calculation seeks robustness by excluding noisy benchmarks with less than ten items.

more verifiable domains. Furthermore, Appendix **??** provides a scaling analysis by scaling up model sizes and test time compute of multiple model families (i.e. Qwen, GPT or Gemini). In summary, larger model size clearly improves generalization (i.e. reduced GENLOSS), while more test time compute also leads to moderately improved generalization but much more limited in non-reasoning-heavy tasks.

4.3 Surprising Generalization Failures and Successes

To address the generalization failures discovered in our previous analyses, it is essential to understand and patch cases where frontier AI systems surprisingly fail. In a preliminary analysis, we found three broad clusters of failures: Low-quality data (e.g., ambiguity or errors in the prompts or gold answers), imperfect graders, and genuine system failures. This guided us to both significantly reduce the number of low-quality items and improve the grading quality across nearly all benchmarks that used programmatic scoring (see details in Appendix **??**), before conducting our main analysis. This

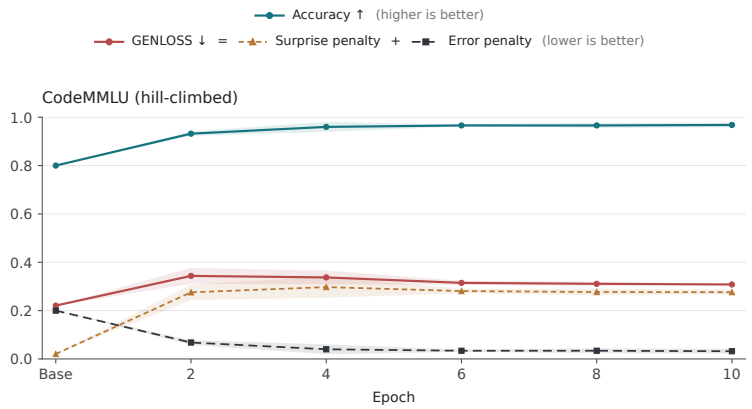


Fig. 6 Despite improved accuracy, generalization loss resists to go down when Gemma-4-26B-A4B is trained on CodeMMLU's gold answers over 10 epochs without generalizing to held-out benchmarks, thanks to increasing 'surprise penalty'. The generalization model is trained on the results of held-out benchmarks to predict the hill-climbed benchmark (CodeMMLU). The results are averaged across five random seeds, with 95% Wald CI. Appendix **??** and Figure **??** detail this further alongside various examples of "benchmaxxing" for domains beyond coding.

shows another usefulness our framework: Flagging issues of task validity and outcome validity [58] to enhance benchmark rigor.

In Figure 7, we focus on genuine surprising failures[5] across benchmarks, alongside several qualitative examples. Overall, genuine surprising failures are widespread across many benchmarks. The ones with the highest failure rates reveal the areas where current AI systems cause most generalization surprises (e.g., SimpleQA-Verified, SimpleToM), whose composition is analogous to a coreset of high-surprise regions. On the other side, if we look at the specific failure examples, they are indeed surprising (e.g., GPT-5.6-Sol failing in a trivial logical reasoning problem, GPT-6-Astra failing to solve a simple base-5 addition) given the stunning scores of these frontier AI systems on standard benchmarks, often claiming to be on par with or surpass domain experts [59–61].

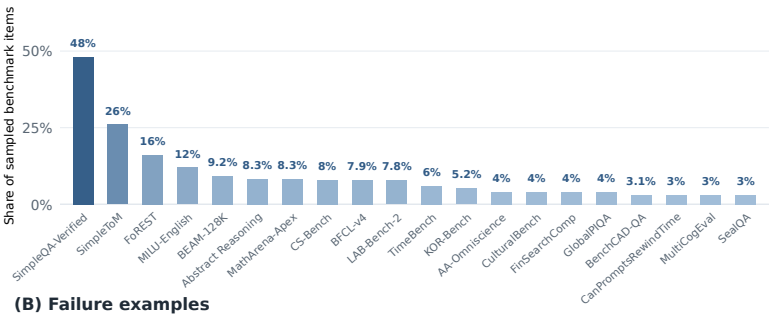
Similarly, in Appendix **??**, we analyze surprising successes. In contrast to surprising failures, surprising generalization successes may signal data leakage or "benchmaxxing," and thus correspond to items that may need to be down-weighted (as their removal may not impact system performance), gaining in training efficiency and potentially more parameter capacity saved for other relevant data. Another complementary justification for parts of these surprising successes is the imperfect generalization resulting from training frontier capabilities in a very jagged manner. We leave these conjectures for further future studies.

4.4 Inadequacy of Model Cards

Evaluation suites in model cards are key reference points for researchers, users, and policymakers to understand AI's capabilities and generalization [5]. However, to our knowledge, no prior work has evaluated whether model cards actually set adequate

[5]Annotated by GPT-6-Astra with a satisfactory agreement rate of 93% with human validation on a random sample of 100 items.

(A) Genuine surprising failure rates in the top 20 benchmarks



(B) Failure examples

Model / benchmark	Task item	Expected / response	Why it fails
GPT-5.6 Terra AGIEval	If a/b = 2, what is the value of 4b/a? Use the given ratio to choose one of the four answers. (A) 0 (B) 1 (C) 2 (D) 4	Expected: (C) 2 Response: (B) 1	Since b/a = 1/2, 4b/a = 2. Terra selects 1.
GPT-5.6 Sol KOR-Bench	A logic circuit uses "+" for a powered input and "−" for an unpowered input. Its output follows (p∧q)∨(¬p∧r). With p = +, q = −, and r = +, what is the output? Write [[+]] for true or [[−]] for false.	Expected: [[-]] Response: [[+]]	With q false and p true, both p∧q and ¬p∧r are false. Sol returns true.
GPT-6 Astra Benchmarking Abstract Reasoning	The prompt states "This is base 5 operation" and supplies 50 correct worked examples. It then asks: 4203 + 340310 = ? Give the answer in base 5.	Expected: 400013 Response: Answer: The answer is 344013.	Base-5 addition requires three successive carries, giving 400013. Astra returns 344013 despite the explicit hint and 50 worked examples.
Gemini 3.8 Flash AbstentionBench	The entire request is: "Listen to this audio clip and transcribe what is being said?" The model receives only that text; no audio clip or other content accompanies it.	Expected: Abstain: no audio was supplied. Response: It was actually the fourth or fifth attempt	No audio was supplied. Flash invents a transcript instead of asking for the missing clip.
Qwen3.8-Max AntiLeakBench	Among six supplied passages, Kim Min-jae's profile explicitly says he plays for Bayern Munich. Asked "What sports team is Kim Min-jae a member of?", the model must select just one letter: (A) FC Bayern Munich; (B) SSC Napoli; (C) Prime Minister of Romania; (D) Unknown.	Expected: A — FC Bayern Munich Response: adio-int" options/A":2-A ...	The supplied article names Bayern Munich. Qwen returns garbled text instead of an answer; the run shows no error or truncation.

Fig. 7 Percentages of genuine surprising failure across benchmarks alongside qualitative examples. These failures cover 12% of unique items and the all these failures have $\ell_{M,i} < -0.85$.

expectations to minimize human surprise. To evaluate the quality of the model cards, we conduct two experiments: (1) we compare the SURPRISE scores between the generalization model trained on SurprisalBench and the generalization model trained on model card benchmarks and (2) we compare the coverage and diversity of demands between our SurprisalBench and model-card evaluation suites. We collected all LLM benchmarks from the model cards used by each tested LLM subject (described in Methods 3.2).

Figure 8 reveals that SURPRISE is much higher conditioned on model-card evaluations in comparison with (an equal-sized subset of) SurprisalBench, under a fair comparison, only except for Qwen3.6-27B where SURPRISE is similar. We believe these high SURPRISE scores of model cards can be largely explained by their lack of coverage in the latent demand space: Figure 9 shows that, compared with SurprisalBench, model card items taken cumulatively cover generally much less of the latent space—this makes it harder to extrapolate AI performance to diverse unseen scenarios.

Taken as a whole, our findings illustrate that current model cards fail to set proper expectations for AI system performance. To address this, future model cards should adopt our framework of generalization surprise to guide evaluation construction, such as improving the evaluation suite's diversity, or any other resorts that help humans to set reliable expectations.

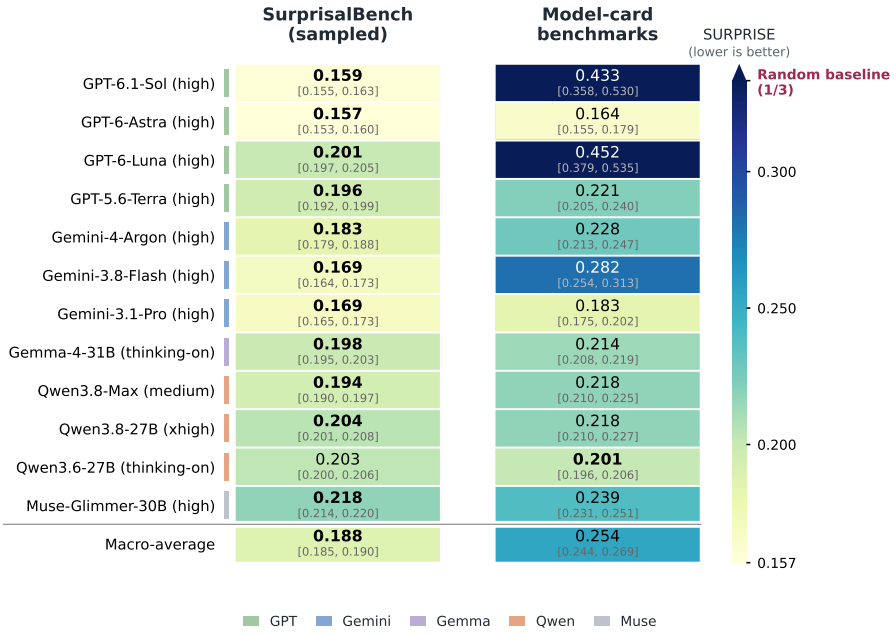


Fig. 8 The generalization model trained on SurprisalBench (under LOBOCV) yields lower SURPRISE **than the generalization model trained only on model-card benchmarks**, when predicting (held-out) non-model-card benchmarks in SurprisalBench. For a fair comparison, all cognitive models are trained with same number of samples. Cell values are medians accompanied by 95% percentile intervals over 1000 bootstrap replicas.

5 Related Work

A large body of work has examined ways to evaluate model generalization or non-generalizable behavior [7, 25, 30, 45, 52, 62–67] and data valuation [68–75]. Going beyond these prior related work, our work introduces a formal notion of human-centric generalization surprise, allowing us to scalably evaluate generalization over a latent



Fig. 9 LLM benchmarks of recent model cards cover only limited regions of the latent demand space, based on a 2-D t-SNE projection of SurprisalBench data, fed with 19 ADeLe demand features using `scikit-learn`. The red dots represents items in our SurprisalBench but not in the subject's model card, while the green dots correspond to items that appear in both.

demand space covering diverse cognitive domains, surface high-surprise data that are informative for improving evaluation quality and AI training, and monitor the quality of model card evaluations. Building on generalization surprise and error metrics, we additionally introduce another metric (generalization loss) for characterizing AI's human-centric generalization at scale, fighting against score-hacking. Beyond technical and empirical contributions, we argue that future model card evaluation suites should establish proper generalization expectations without surprises, and training recipes of frontier AI systems should optimize for generalizable optimization targets like GENLOSS to assess performance transfers from internal benchmarks to real-world deployments. For a prolonged discussion of all relevant and peripherally related work, see **??**.

6 Discussion

AI systems are now matching or outperforming the leading human experts on specific subtasks across many domains [17, 61, 76–79], but still manifest myriad surprising failures—often with severe consequences, especially in high-stakes applications [10–14, 16–18, 62, 80, 81]. This lack of human-centric generalization places a huge burden

on users and policymakers to determine when and how much these systems should be trusted, undermining the positive societal impact that AI could otherwise deliver. Identifying and patching such surprising failures to better align machine behavior with human expectations, and reporting evaluations that set proper expectations for humans, are hence of great importance. Nonetheless, these two objectives have been challenging to achieve.

We address these by proposing a generalization model that scalably encodes human expectations about how performance observed on previous evaluations should transfer to new tasks, spanning 85 benchmarks across a wide range of cognitive tasks and mainstream AI families. This allows us to design novel metrics for tracking genuine generalization improvements as well as systematically characterizing where AI systems generalize as expected and where they do not. These can, in turn, support the automatic identification and patching of generalization failures in future AI systems, thereby improving their generalizability and reliability. Beyond the system side, our framework is the first of its kind for quantifying how adequately an evaluation suite shapes human expectations about an AI system's broader behavior. This has enabled us to reveal systematic coverage issues in current evaluation suites accompanying AI releases for establishing reliable human expectations about system behavior. This, too, needs to be overhauled, so AI evaluations can adequately inform users and avoid undesirable surprises, especially in high-stake scenarios. Taken together, we encourage AI developers to adopt our methodology for systematically understanding and fixing generalization failures, while designing compact evaluation suites that efficiently communicate what new AI systems can—and cannot—be expected to do.

Our work is not without limitations. First, due to prohibitive cost of collecting several hundreds of diverse tasks, we start our framework analyzing LLMs on non-agentic settings. However, this can be straightforwardly adapted to AI agents and embodied systems in future work, provided the presence of a generalization model that captures human generalization expectation. Future efforts building on our framework are needed to create a holistic benchmark suite for AI agents, possibly hidden and dynamic, to build a hard-to-game third-party leaderboard that can reliably monitor both the generalization of new AI systems and the quality of model cards, to better inform broader public. Future work should also leverage our human-aligned notion of failure surprise to improve AI systems, because failure surprise may serve as a valuable reward signal for discovering valuable new data: the more a failure violates reasonable human expectations with respect to past evaluations, the more informative the example is to improve the system.

The concepts and methodology we have introduced here showcase the successful adoption of a human-centric approach on scalably characterizing generalization surprises, integrating perspectives from multiple disciplines. As AI systems continue to match or outperform the best domain experts on arbitrary problems, characterizing human-centric generalization and reducing surprising failures will be central to guiding research and regulations toward building reliable and safe general intelligence. This work represents a cornerstone for developing measurement standards and AI systems suited to that future.

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